

MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU

SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Scheme of Evaluation Cloud Computing [CSE_4063]

1. a) A large university, GlobalTech University, plans to implement a cloud-based Learning Management System (LMS) for students and faculty. The system will store student records, assignments, and research data. The university wants to:

- Ensure data security and privacy for sensitive academic information,
- Allow scalability to handle thousands of users during exams or admissions,
- Facilitate collaboration with other universities for joint research projects, and
- Keep operational costs under control without maintaining too much hardware.

As a Cloud Computing Consultant, recommend the most appropriate cloud deployment model (Public, Private, Community, or Hybrid) for GlobalTech University. Apply the features of the chosen model to justify how it meets the university's requirements.

Answer: Recommended Deployment Model: ✓ Hybrid Cloud

GlobalTech University should adopt a **Hybrid Cloud** deployment model because it provides the right balance of **security**, **scalability**, **collaboration**, and **cost efficiency** needed for an academic institution handling sensitive and high-volume data.

1. Ensuring Data Security and Privacy

The university can store sensitive information—such as student records, grades, exam data, and confidential research—on the **private cloud** portion.

- Private cloud offers **high security**, **strict access control**, and **compliance** with academic and data protection policies.
- The university retains full control over critical data while still using cloud benefits.

Why it fits: Protects sensitive academic data without exposing it to external public infrastructure.

2. Scalability for Thousands of Users

During admissions, exams, or LMS peak usage, workload drastically increases.

- The **public cloud** component of the hybrid model allows **on-demand scalability** to handle thousands of concurrent learners and faculty members.
- Resources expand automatically and contract when load reduces.

Why it fits: Supports huge, temporary spikes without requiring permanent hardware purchases.

3. Collaboration with Other Universities

Joint research projects often require secure, shared platforms.

- Hybrid cloud makes it easy to share **non-sensitive or anonymized research datasets** through the **public cloud**, enabling smooth collaboration.
- Meanwhile, confidential data and ongoing research stay protected in the private cloud.

Why it fits: Enables academic partnerships without compromising data integrity.

4. Cost Efficiency and Reduced Hardware Maintenance

The university avoids large investments in physical servers.

- Public cloud reduces operational costs through **pay-as-you-go** pricing.

- Only critical systems stay on private cloud, reducing overall infrastructure maintenance.
- Hybrid model prevents over-provisioning, which is common in fully private setups.

Why it fits: Offers the right blend of cost savings and control.

Q 1 b) A small digital marketing company, AdSphere Media, is planning to launch a new web application that provides real-time campaign analytics for its clients. The company's goals are to:

- **Develop and deploy the application quickly without managing complex infrastructure,**
- **Use ready-made software tools for email marketing and data visualization,**
- **Scale resources automatically as the number of users increases, and**
- **Keep overall IT maintenance and cost low.**

As a Cloud Computing Consultant, suggest the most suitable cloud service model(s) (IaaS, PaaS, or SaaS) for AdSphere Media. Apply the features of the chosen model(s) to justify how they address the company's needs.

Answer: Recommended Cloud Service Models: ✓ PaaS + SaaS (with optional minimal IaaS)

AdSphere Media should primarily use Platform as a Service (PaaS) and Software as a Service (SaaS) to meet their needs, as these models reduce infrastructure management, support fast development, and keep costs low.

1. Platform as a Service (PaaS) — Best for application development & deployment

Why PaaS?

PaaS provides a fully managed environment with built-in tools for developing, testing, and deploying applications.

How PaaS meets AdSphere's needs

- **Quick development and deployment:**

PaaS offers pre-configured environments, databases, runtimes, and DevOps tools, enabling the team to focus purely on writing code instead of configuring servers.

- **No infrastructure management:**

The cloud provider handles OS updates, middleware, server provisioning, load balancing, and database administration.

- **Automatic scaling:**

When more clients access real-time analytics, PaaS automatically scales CPU, memory, and instances to maintain smooth performance.

Result: Faster product launch with minimal technical overhead.

- **2. Software as a Service (SaaS) — Best for ready-made marketing tools & visualization**

- **Why SaaS?**

SaaS provides ready-to-use software that can be accessed over the internet without installation.

- **How SaaS meets AdSphere's needs**

- **Ready-made tools for email marketing:**

Services like Mailchimp, SendGrid, or HubSpot allow the company to run campaign emails without building a custom tool.

- **Pre-built data visualization and analytics apps:**

Tools like Google Analytics, Power BI, or Tableau Online help create dashboards instantly without custom coding.

- **Low cost & zero maintenance:**

SaaS eliminates the need for updates, patches, or hardware—everything is handled by the vendor.

Result: Faster operations with reduced development effort and cost.

3. (Optional) Infrastructure as a Service (IaaS) — Only if custom components are needed

Where IaaS may help

If AdSphere needs custom virtual machines for proprietary analytics engines or storage systems, IaaS can be used for isolated components.

Why it is optional

Since the company wants minimal infrastructure management, IaaS should be used sparingly.

Q 2 a) A public cloud service provider needs to ensure data isolation while provisioning virtual machines to different tenants.

a) Explain how virtualization ensures logical isolation among multi-tenant environments.

b) Propose additional measures to strengthen VM security and tenant isolation.

Answer:

a) How virtualization ensures logical isolation in multi-tenant environments

Virtualization provides logical isolation through several mechanisms:

- **Hypervisor-based abstraction:**
Each Virtual Machine (VM) runs on top of a hypervisor (Type-1 or Type-2), which creates separate virtual hardware environments. Tenants cannot directly interact with the underlying physical hardware or other tenants' VMs.
 - **Virtual Memory Isolation:**
The hypervisor maintains independent memory spaces for each VM using techniques such as shadow page tables, nested paging, and Extended Page Tables (EPT). One VM cannot read or modify another VM's memory.
 - **CPU Scheduling Isolation:**
Hypervisors use virtual CPUs (vCPUs) mapped to physical CPUs. Scheduling ensures one VM's workload does not interfere with another tenant's execution context.
 - **Network Isolation:**
Virtual networks (VLANs/VXLANs), virtual switches, and SDN ensure isolation of traffic flows. Packets belonging to one tenant cannot reach another tenant's VM without explicit routing.
 - **Storage Isolation:**
Each tenant receives a logically separated virtual disk image. Access control and file-system virtualization prevent cross-tenant visibility.
 - **Privilege Separation:**
VMs run in unprivileged guest mode, while the hypervisor operates in a highly privileged mode. This prevents VMs from executing operations that affect other tenants or the host.
- Overall, virtualization ensures logical isolation by partitioning compute, memory, storage, and networking resources so tenants behave as if they have dedicated infrastructure.

b) Additional measures to strengthen VM security and tenant isolation

To enhance security beyond basic hypervisor isolation, cloud providers can implement:

1. Hardware-Assisted Security

- Use Intel VT-x, AMD-V, Intel VT-d, and IOMMU to prevent DMA attacks.
- Employ Trusted Platform Module (TPM) and secure boot for VM attestation.

2. Strong Access Control and Identity Management

- Implement Role-Based Access Control (RBAC) for administrators.
- Enforce MFA and audit logging for all management operations.

3. Network Micro-Segmentation

- Use SDN-based policies, firewalls, and Zero-Trust network segmentation.

- Apply East–West traffic filtering between VMs.
4. VM Image Security
- Scan VM images for malware and vulnerabilities before provisioning.
 - Sign and validate VM templates to prevent tampering.
5. Runtime Security
- Install hypervisor-level intrusion detection/prevention systems (HIDS/HIPS).
 - Use agent-based VM behavior monitoring (anti-malware, activity profiling).
6. Resource Hardening
- Limit and monitor VM privilege escalation attempts.
 - Restrict sharing of devices such as GPUs, NICs, and storage controllers.
7. Protect Against Side-Channel Attacks
- Use cache partitioning and flush techniques to isolate CPU cache lines.
 - Disable simultaneous multithreading (SMT) where required (e.g., for cryptographic tenants).
8. Encrypt Data Everywhere
- Full-disk encryption for VM volumes.
 - Encrypt data-in-transit using TLS/IPsec between virtual networks.
 - Provide Confidential Computing (Intel SGX, AMD SEV).

Q 2 b) In a hybrid cloud, workloads are inconsistently balanced across VMs.

a) Analyze the resource allocation metrics that lead to imbalance.

b) Evaluate how dynamic VM migration can improve overall load distribution.

Answer: a) Analysis of Resource Allocation Metrics Leading to Imbalance

In a hybrid cloud environment, workload imbalance across virtual machines occurs when key resource metrics are unevenly consumed or provisioned. The main metrics contributing to imbalance include:

•CPU Utilization:

Unequal distribution of CPU-intensive workloads causes some VMs to reach high CPU saturation while others remain underutilized.

•Memory Usage:

Poor memory allocation or sudden spikes in memory-heavy tasks lead to swapping, reduced performance, and imbalance across VMs.

•Disk I/O Throughput:

VMs running storage-intensive applications can create bottlenecks, causing significant latency differences compared to lightly loaded VMs.

•Network Bandwidth Consumption:

Excessive east–west or north–south traffic on specific VMs leads to congestion, while others may have unused bandwidth.

•Application-Level Metrics:

Variations in request rates, session loads, or microservice invocation patterns can overload specific VMs.

•Static Resource Allocation Policies:

When fixed resource reservations or rigid load-distribution rules are used, they fail to adapt to varying workload types, causing persistent imbalance.

These metrics collectively determine how evenly workloads are distributed across the hybrid cloud; imbalance arises when one or more metrics deviate significantly across VMs.

b) Evaluation of How Dynamic VM Migration Improves Load Distribution

Dynamic VM migration allows running workloads (VMs) to move across physical hosts or between private and public cloud segments in a hybrid cloud. It improves load distribution through:

•Real-Time Load Redistribution:

Overloaded hosts can offload VMs to lightly loaded hosts, smoothing CPU, memory, and I/O utilization across the environment.

- **Optimized Resource Utilization:**

Migration ensures that no single host becomes a performance hotspot; resources across the hybrid cloud are used more uniformly.

- **Improved Application Performance:**

Migrating VMs away from congested nodes reduces latency, enhances throughput, and prevents SLA violations.

- **Energy Efficiency:**

Underutilized hosts can be consolidated by migrating VMs off them—allowing idle hosts to be powered down or enter low-energy states.

- **Fault Tolerance and Resilience:**

If a host shows signs of failure or instability, VMs can migrate preemptively, preventing disruptions.

- **Elastic Expansion to Public Cloud:**

When private cloud resources are saturated, migration to public cloud nodes ensures continuous scaling without performance degradation.

Thus, dynamic VM migration provides an adaptive mechanism to continuously balance load, minimize hotspots, improve responsiveness, and enhance the operational efficiency of hybrid cloud infrastructures.

Q 3 a) A smart city uses a Message Bus with a Publish–Subscribe model to deliver traffic and pollution alerts to citizen apps.

a) Analyze how does the message bus reduce direct dependencies between services in this system?

b) Which filtering method (topic-based or content-based) would you apply for sending only pollution alerts to subscribed users? Justify.

c) Explain how event-driven alerts help city authorities take immediate action.

Answer: a) In a Publish–Subscribe architecture, the Message Bus acts as an intermediary between publishers (traffic sensors, pollution sensors, servers) and subscribers (citizen apps).

It reduces direct dependencies in the following ways:

- Publishers do not need to know who the subscribers are or how many exist. They simply send messages to the bus.
- Subscribers do not communicate directly with publishers. They receive messages from the bus based on their subscriptions.
- The message bus handles routing, delivery, transformation, and communication, so sensors or apps do not depend on each other's protocols or implementation details.
- Changes in one service do not affect others.

For example, updating pollution sensor software does not require modifying citizen apps.

b) Recommended Method: Topic-Based Filtering

Justification:

- Pollution alerts can be categorized under a specific topic such as "pollution.alerts" or "airquality.pm25".
- Subscribers who want only pollution data simply subscribe to that topic.
- It is simple, efficient, and scalable because sensors continuously generate high-volume alerts.

Why not content-based?

- Content-based filtering evaluates message payloads (e.g., checking PM2.5 levels), which introduces processing overhead.
- Topic-based filtering works better for real-time, city-wide systems where rapid delivery is needed.

c) Event-driven alerts notify authorities as soon as a significant event occurs, such as:

- Pollution threshold crossing
- Sudden traffic congestion
- Accident or road blockage

This helps authorities by:

- Real-time decision-making:

Alerts are triggered instantly, enabling quicker responses.

- Automated workflow triggering:

High pollution → trigger air-quality inspection teams

Heavy traffic → trigger dynamic traffic light adjustments or rerouting

- Faster public communication:

Authorities can push advisory messages (e.g., "Avoid Area A due to high PM2.5").

- Better resource deployment:

Emergency teams, traffic police, or environmental units can be dispatched immediately.

Q 3b) A research lab runs a complex climate simulation workflow on a distributed grid system. The workflow includes sequential preprocessing tasks and parallel computation tasks that require frequent data sharing across nodes.

a) Analyze how a DAG-based workflow architecture ensures correct task execution and efficient resource usage for this simulation.

b) Explain how the workflow execution engine handles control flow vs. data flow and ensures fault tolerance during long-running scientific computations.

Answer: a) A Directed Acyclic Graph (DAG) represents the climate simulation workflow as tasks (nodes) with dependencies (edges). It ensures:

- Correct task ordering – Preprocessing tasks appear as parents in the DAG, so the system guarantees they finish before parallel climate computation begins. No circular dependencies ensure error-free execution.
 - Parallelism where possible – Independent computation tasks are identified automatically in the DAG, enabling parallel execution across multiple grid nodes, improving throughput.
 - Efficient resource allocation – The scheduler uses the DAG structure to allocate HPC or grid resources only when a task becomes ready, preventing idle nodes and reducing overall execution time.
- The DAG ensures accurate sequencing of dependent tasks and maximizes parallel resource usage across the distributed grid.

b) Control flow vs. data flow handling & fault tolerance by the execution engine:

1. Control Flow Management

- The workflow engine enforces the logical order defined in the DAG (task A → task B → task C).
- It triggers tasks only when all prerequisite tasks are marked as complete.
- This guarantees correct coordination of sequential preprocessing and parallel computation stages.

2. Data Flow Management

- The engine manages data staging, transfers, and sharing between nodes during simulation.
- Intermediate outputs from preprocessing tasks are automatically routed as inputs to the next stage.
- Caches or temporary storage minimize repeated data movement across nodes.

3. Fault Tolerance

If a grid node fails during a long-running climate computation, the engine supports:

- Checkpointing (saving intermediate states),

- Task retry on another node,
 - Re-execution of failed tasks without restarting the whole workflow.
- This ensures high reliability for multi-hour or multi-day scientific simulations.

4 a) A startup is developing a real-time food delivery tracking system. The application requires automatic scaling during peak meal times, fast deployment, and event-driven functions for processing live location updates.

a) Analyze how Google App Engine's Standard vs Flexible environments differ in supporting this application's scaling, runtime control, and execution requirements.

b) Evaluate whether AWS (Lambda, Elastic Beanstalk) or Microsoft Azure (Functions, App Services) provides a better programming environment for event-driven processing and rapid integration with backend storage — justify with service features.

Answer: a) Google App Engine Standard

- Scaling: Very fast automatic scaling (instances start quickly → ideal for peak meal-time spikes).
- Runtime Control: Limited control; uses predefined, sandboxed runtimes.
- Execution Suitability: Best for lightweight, request-driven tasks like handling location updates with low latency.

Google App Engine Flexible

- Scaling: Slower scaling; relies on Docker containers and VM-like instances.
- Runtime Control: Full flexibility — custom runtimes, libraries, background processes allowed.
- Execution Suitability: Good for long-running tasks or custom frameworks but less ideal for rapid auto-scaling required by real-time tracking.

b) AWS (Lambda + Elastic Beanstalk)

- Lambda: Strong event-driven model, triggers from S3, DynamoDB, API Gateway.
- Elastic Beanstalk: Easy deployment but less event-focused.

Azure (Functions + App Services)

- Functions: Excellent for real-time triggers from Cosmos DB, Blob Storage, Event Hub.
- App Services: Smooth integration with Azure Storage and databases.

Azure Functions generally offers tighter, built-in integration with backend storage and real-time events, making it slightly better for location-update processing. AWS Lambda is powerful, but Azure provides more seamless end-to-end event and storage workflows.

Q 4 b) A cloud provider offers three kinds of agreements: one for individual users, one for enterprise customers, and one between internal departments of the same organization. Analyze your understanding of Service Level Agreements (SLAs) to identify the type of SLA represented in each case and explain how their scope and application differ in real-world cloud service scenarios.

Answer:

1. Agreement for Individual Users – Customer-based SLA

- This SLA is designed for a specific individual customer.
- It defines the services, performance guarantees, and support provided to that single user.
- Common in public cloud services where users subscribe to storage, compute, or applications.

2. Agreement for Enterprise Customers – Service-based SLA

- This SLA is offered to all enterprise clients using the same service.
- The terms are standardized (e.g., 99.9% uptime, backup schedules) regardless of the specific enterprise.
- Ensures consistency and uniformity for businesses using the same cloud solution.

3. Agreement between Internal Departments – Multi-level/Corporate-level SLA

- a. Used within the same organization where the IT/cloud team provides services to other internal departments.
- b. Defines internal service responsibilities, performance targets, and compliance requirements.
- c. Helps manage internal operations, governance, and resource allocation.
- 4. Scope Differences
 - a. Customer-based SLAs focus on tailoring needs for a single user.
 - b. Service-based SLAs cover a broad group of users using the same service.
 - c. Corporate/multi-level SLAs focus on internal service quality within the organization.
- 5. Real-world Application
 - a. Public cloud providers use customer-based SLAs for individual plans (e.g., basic user accounts).
 - b. Enterprises get service-based SLAs for large-scale services (e.g., IaaS, SaaS packages).
 - c. Internal IT departments use multi-level SLAs to ensure smooth internal operations and meet organizational standards.

Q 5 a) Apply your knowledge of cloud security mechanisms to explain how encryption, strong authentication, and access control work together to protect sensitive customer data stored on a public cloud platform.

Answer: When sensitive customer data is stored on a public cloud platform, different cloud security mechanisms work together to provide strong protection.

Encryption ensures that data is converted into an unreadable format so that even if someone gains unauthorised access to the storage or intercepts the data in transit, they cannot understand or misuse it without the decryption key. This maintains confidentiality of the customer information.

Strong authentication adds another protective layer by ensuring that only genuine and verified users can access the cloud environment. Methods like multi-factor authentication require users to provide multiple proofs of identity, such as passwords, OTPs, or biometrics. This prevents attackers from entering the system even if they manage to steal login credentials.

Access control further strengthens security by defining what each authenticated user is allowed to do. Using role-based access and cloud IAM policies, the organisation can limit user permissions so that only authorised individuals can view or modify sensitive customer data. This ensures that users get only the minimum privileges needed for their role.

Together, these mechanisms provide layered protection. Encryption safeguards the data itself, strong authentication verifies the identity of users trying to access the cloud, and access control restricts their actions once they are inside. By working in combination, these mechanisms ensure that customer data stored on a public cloud platform remains secure, confidential, and protected from both external and internal threats.

Q 5 b) A financial company uses a multi-layer cloud setup that includes firewalls, encryption, intrusion detection systems, and secure APIs to protect data across different service layers.

Explain how this setup reflects the concept of cloud computing security architecture and how layered protection enhances overall security.

Answer: The given scenario reflects the concept of **cloud computing security architecture**, where security is applied in multiple layers across the cloud environment. The financial company uses different security tools such as firewalls, encryption, intrusion detection systems, and secure APIs, and each of these tools protects a specific layer of the cloud. This shows that cloud security is not dependent on a single mechanism but is designed as a structured architecture where every layer network, data, application, and interface has its own security controls.

The use of **firewalls** represents protection at the network layer by filtering unwanted traffic. **Encryption** protects the data layer by ensuring that even if data is accessed, it cannot be read without the proper keys. **Intrusion detection systems** provide monitoring and alerting at the system level, helping detect suspicious behaviour or attacks. **Secure APIs** ensure that application-level interactions are protected so that only trusted applications and services can communicate with each other.

This multi-layer setup enhances overall security because if one layer fails, the other layers still provide protection. Attackers must get through multiple independent barriers, making it much harder to compromise the system. Layered security also ensures that threats targeting different levels such as network attacks, data breaches, or API misuse are handled by appropriate security tools. Thus, layered protection strengthens the overall defence of the cloud environment and ensures that sensitive financial data remains safe from a wide range of cyber threats.